

Diabetic Foot Complication Avoidance Through a Wearable Sensor and Random Forest Classifier for Automated Evaluation

Arunkumar P

Department of Biomedical
Engineering
KPR Institute of Engineering and
Technology
Coimbatore, India
panarun5@gmail.com

Gayathri KS

Department of Biomedical
Engineering
KPR Institute of Engineering and
Technology
Coimbatore, India
gayathrisivaraj@ieee.org

Ridana Sri S

Department of Biomedical
Engineering
KPR Institute of Engineering and
Technology
Coimbatore, India
ridana07@ieee.org

Bharathi Prabha J

Department of Biomedical
Engineering
KPR Institute of Engineering and
Technology
Coimbatore, India
bharathi06@ieee.org

Thangaraj KN

Department of Diabetology
Balaclinic & Diabetes Centre
Tiruppur, India
knthangaraj@hotmail.com

Deepak D

Department of Biomedical
Engineering
KPR Institute of Engineering and
Technology
Coimbatore, India
deepak10durai@gmail.com

Abstract—Limited early detection and lack of real-time monitoring in Diabetic Foot Ulcer (DFU) management contribute to delayed diagnoses, amputations, and increased healthcare burden, necessitating a paradigm shift towards early detection through non-invasive sensor technology. The wearable solution for DFU detection utilizes Foot Regional Saturation of Oxygen (RSO2), thin-film Foot Temperature Sensors (FTS), and Foot Pressure Sensors (FPS). The system is based on the integration of the Internet of Things (IoT) and the Random Forest Classification algorithm, signifying a paradigm shift in DFU management. The methodology involves the four-layered system architecture that comprises data collections through thin-film sensors, that are processed using machine-learning algorithms for foot ulcer detection. The performance of the proposed Random Forest (RF) classification model has been evaluated using various metrics, with Accuracy (90.02%), Precision (90%), Recall (90.5%), and F1-Score (89.7%). Furthermore, the system is capable of providing diagnosis and personalized health recommendations to assist patients with Diabetic foot ulcers in enhancing their general health and well-being. The system has the potential to enhance the quality of life for individuals with diabetes, representing a significant breakthrough in medical technology, expanding the understanding of diabetic foot ulcer detection, and motivating additional research in this vital area.

Keywords—Diabetic Foot Ulcers (DFUs), Early detection, Continuous surveillance, Wearable solution, Foot Regional Oxygen Saturation (Foot RSO2), Foot Temperature - thin film sensor, Foot pressure sensor, Machine learning

I. INTRODUCTION

Diabetic foot ulcer is a formidable health concern which has a substantial impact on morbidity and mortality rates among individuals with diabetes. Current DFU management methods often rely on visual inspection and physical palpation, which are not always accurate or reliable. There is a need for more accurate and reliable

methods for detecting and monitoring DFUs. The importance of early detection and sustained monitoring for effectively managing these ulcers is unequivocal. The work introduces a pioneering approach that converges the parameters such as Foot Regional saturation of Oxygen (RSO2) measurement capabilities, temperature thin film sensor, and foot pressure sensor for real-time monitoring and early detection of diabetic foot ulcers as illustrated in Fig 1. Globally, the World Health Organization has reported a staggering 422 million people afflicted with diabetes, a figure that has quadrupled since 1980. Among many complications of diabetes, foot ulcers have emerged as a prevalent and severe consequence, affecting up to 25% of diabetic patients over their lifetime [1]. These ulcers pose a grave risk, often culminating in severe infections and dire circumstances, necessitating amputation.

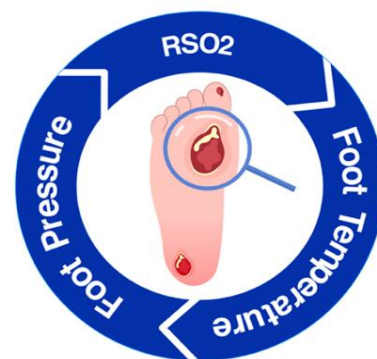


Fig. 1. Overview of a Diabetic Foot Ulcer detection parameters.

The ingenuity of the proposed approach lies in the fusion of three pivotal components: a Foot RSO2 sensor, a thin-film sensor dedicated to temperature measurement, and a foot pressure sensor. These elements are seamlessly integrated into a wearable device to ensure ease of use for the patient. The real-time transmission of data facilitated by

the IoT propels the collected sensor data to a machine-learning model. The synergy of machine learning and IoT augments the efficacy of the approach, transforming it into a potent tool for diabetic foot care. Real-time monitoring facilitates immediate intervention, thereby arresting the progression of ulcers. Rigorous testing on a cohort of patients with diabetes yielded promising results, indicating a high accuracy rate in the early detection of foot ulcers.

The amalgamation of Regional Saturation of Oxygen (RSO2), thin-film Foot Temperature Sensors (FTS), and Foot Pressure Sensors (FPS) within a wearable device [2] [17] is a promising solution for the timely detection of diabetic foot ulcers is illustrated in Fig 1. The DFU detection system uses a four-layered system architecture integration of the Internet of Things (IoT) and the Random Forest Classification algorithm offers a promising solution for non-invasive, real-time monitoring of foot health, enabling early intervention and improved patient outcomes. The advent of such a system has the potential to revolutionize the landscape of diabetic foot ulcer detection and management, providing an opportunity to save many lives and significantly curtail healthcare costs.

II. PROPOSED METHODOLOGY

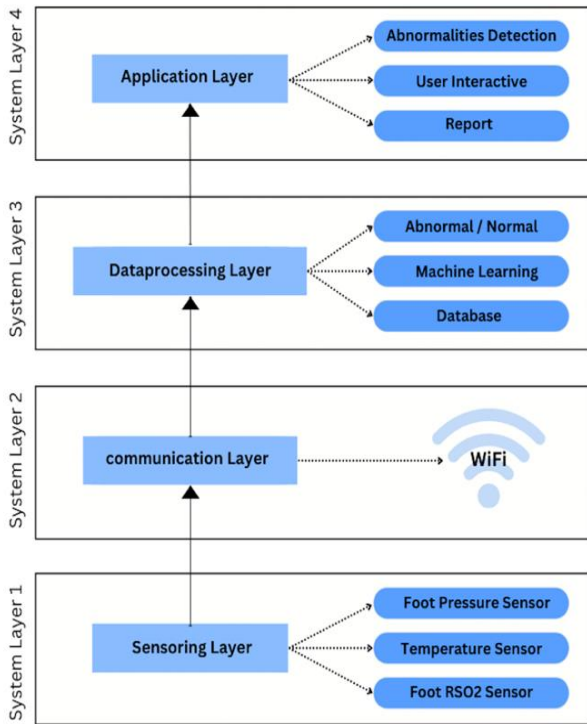


Fig. 2. System Architecture - Diabetic Foot Ulcer Detection.

The approach is focused on the development of a real-time, wearable device for detecting diabetic foot ulcers [2][17]. The device incorporates a four-layered system architecture with three types of sensors: a Foot Pressure Sensor [18], a Negative Thermal Coefficient (NTC) Thin Film Thermistor for measuring temperature, and an RSO2 Sensor for monitoring oxygen saturation in the foot [4]. These sensors provide essential information about the foot's condition, which is critical for early detection of foot ulcers. The data collected from these sensors is used to calculate shear stress, temperature, and oxygen saturation on the foot,

a key factor in the development of foot ulcers. The project uses an ESP32 and WiFi module for communication between the sensors and microcontroller. The microcontroller processes the sensor data and employs a machine-learning model for detecting foot ulcers. The project offers real-time monitoring of the foot's condition, alerting the user and healthcare provider when a potential foot ulcer is detected. The methodology represents a significant advancement in diabetic foot ulcer detection, providing a wearable solution [17] that integrates IoT and machine-learning technologies [3]. The architectural elements are shown in Fig 2.

A. System Layer I

The Diabetic Foot Ulcer Detection methodology relies on specific healthcare parameters that utilize various sensors, including the foot pressure sensor, which employs force-sensing resistors (FSR) to transform pressure into analogue voltage signals based on Ohm's law [6][18]. The Foot tissue oximeter sensor determines the Regional saturation of Oxygen (RSO2) at the foot [5]. Conversely, the NTC thin-film thermistor sensor translates output voltage directly into corresponding temperature values [16]. These stages involve the physical attachment of sensors to the ESP32 Module and ongoing data collection from these sensors. The parameters are computed using specific equations:

$$FPA = \sum_{i=1}^8 (FP_i - FPN) \quad (1)$$

$$TA = \sum_{i=1}^6 (Ti - TN) \quad (2)$$

$$RSA = \sum_{i=1}^3 (Ri - RN) \quad (3)$$

Fig 3 illustrates the direct proportionality between the pressure applied and the output voltage in a foot pressure sensor. As the pressure increases, the output voltage also increases linearly, highlighting the sensor's sensitivity and precision.

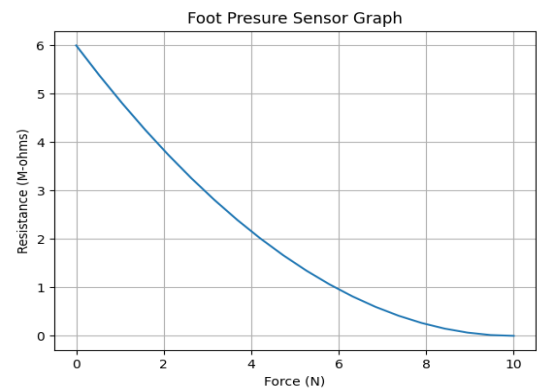


Fig. 3. Foot Pressure Sensor: Force vs. Resistance Graph.

Fig 4 represents the fluctuations in oxygen saturation levels over time. The blue line, labelled as the RSO2 signal, oscillates between 83 and 87, offering insights into the dynamic nature of oxygen saturation in the human foot [7].

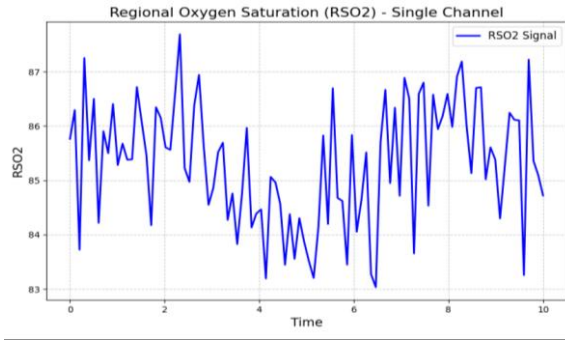


Fig. 4. RSO2 Sensor Data: Fluctuation Over Time.

Fig 5 illustrates the dynamic changes in foot temperature, measured in Celsius, over 10 seconds. The intricate oscillations captured in the data underscore the complexity of thermal regulation in the human foot.

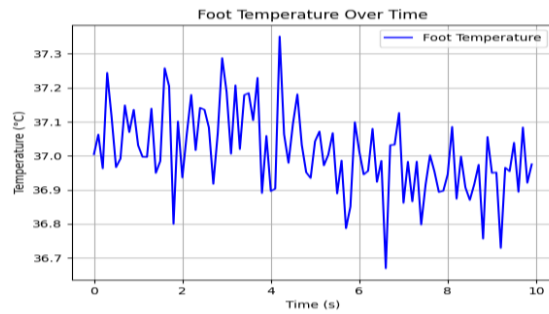


Fig. 5. Foot Temperature – NTC Thin Film thermistor Data: Fluctuation Over Time.

These individual parameters are then utilized to calculate the Diabetic Foot Ulcer Assessment Index (DAI) as the primary metric for quantifying DFU risk in real-time. The DAI integrates data from sensors that are attached to the insole that provide foot pressure, temperature, and RSO2 data through a weighted formula :

$$DAI = W_1 \sum (FP_i - FPN) + W_2 \sum (T_i - TN) + W_3 \sum (R_i - RN) + B \quad (4)$$

TABLE I: DiabeticFoot Ulcer Assessment Index (DAI) Calculation

Sensor Modality	Value (i)	Normal Range (N)	Difference (Vi - N)	Weight (Wi)	Weighted Difference (Wi * (Vi - N))	Condition (Higher = Worse)
Foot Pressure (kPa)	65	25-50	15	0.6	9	Elevated pressure
Foot Temperature (kPa)	1.5	0-3	1.5	0.3	0.45	Increased temperature
RSO2 (%)	65	65-80	-5	0.1	-0.5	Decreased oxygenation
Bias	0.2	N/A	N/A	N/A	0.2	N/A
DAI	9.7	N/A	N/A	N/A	N/A	High DFU risk

This table indicates the Diabetic Foot Ulcer Assessment Index (DAI) score of 9.7 indicates a severe ulcer based on elevated pressure, increased temperature, and decreased oxygenation.

The chosen weights reflect the relative importance of each sensor modality in DFU prediction. The calculated DAI score provides a single, interpretable risk indicator of Diabetic Foot Ulcer condition. The Diabetic Foot Ulcer Assessment Index Calculation of an individual is tabulated in Table I.

To interpret, a threshold for the total DAI is established. The threshold is adjusted based on specific sensitivity and specificity requirements, allowing the system to be fine-tuned for the desired health assessment process. The threshold acts as a decisive point, categorizing individuals into distinct interpretations, such as "**Normal Foot**" or "**Abnormal Foot**." By modifying the threshold, the system balances accurately identifying individuals requiring further attention with minimizing false alarms. The approach enables a flexible and adaptable DFU assessment, adapting to various priorities and considerations within the analysing process.

B. System Layer III

In the second layer of system architecture, encompass the communication module - ESP32 WiFi module continuously acquires analog voltage signals from foot pressure, temperature, and RSO2 sensors [8]. The ESP32 then digitizes the data and wirelessly transmits it to the microcontroller server using WiFi. The server stores the data in a database and deploys machine learning algorithms to analyze the data, determine the Diabetic Foot Ulcer Assessment Index (DAI), and categorize foot ulcers. The server also sends risk alerts to patients and clinicians through a user interface when the DAI threshold indicates the potential for ulcer formation. Moreover, the system prioritizes data privacy through layered security, including encryption, access control, and exploring federated learning to keep data on-device, ensuring responsible stewardship of sensitive health information.

C. System Layer III

In the third layer of our system's architecture, we utilize the Random Forest model algorithm for real-time data classification. The model is particularly adept at accurately differentiating between normal and abnormal DFU conditions, thereby significantly improving the accuracy and efficiency of health evaluations [9].

1) *Database and DFU Monitoring and Control:* The key component of the system is that efficiently stores and retrieves sensor data from the smart insole. It is a reliable repository for historical records and real-time monitoring of the DFU condition. The database contains a comprehensive dataset of 5,546 individuals equally divided between patients and healthy subjects. The dataset helps in testing, validation, and identification. It comprises sensor readings that have been meticulously labelled according to established thresholds.

The optimal adjustment of the kernel bandwidths between visualization and evaluation can lead to the maximization of diagnostic insight. By expanding the visualization width to 0.8, the true form of the distribution can be accurately assessed. Reducing the visualization width to 0.2 for testing allows for performance measurement while preserving the differences. The distribution of the target variable (Normality/Abnormality) in both the training and testing datasets is depicted in Fig 6. The bar chart displays a balanced distribution across both sets, ensuring reliable model performance.

The system employs a highly targeted calibration paradigm that effectively harnesses the graphic potential of kernels. It optimizes visualization clarity and prioritizes rigorous evaluation, effectively exposing relevant signals within the noise and facilitating reliable analytics.



Fig. 6. Target Variable Distribution in Training and Testing Data.

Each data point is labeled based on a predefined DFU risk threshold, acting as a critical reference point for classifying foot health. Values falling within the threshold are labeled as "*Normal*" while those exceeding it are designated as "*Abnormal*" streamlining DFU detection [9].

2) *Machine Learning (RF):* The machine learning algorithm Random Forest is applied to the DFU dataset and demonstrates exceptional performance in diabetic foot ulcer prediction, achieving 93% accuracy and 92.5% precision through feature selection process which is illustrated in Fig 7. Such high accuracy and precision reveal Random Forest's strength in correctly categorizing data points as diabetic foot ulcers or non-ulcers while maintaining a balance between true positives and negatives [13]. The high degree of precision enables Random Forest to make accurate diabetic foot ulcer predictions that can drive preventative care [15]. Overall, these robust metrics confirm the reliability of the Random Forest algorithm for maintaining foot health by enabling early detection of impending ulcers based on real-time multimodal data. The

ensemble approach allows Random Forest to effectively learn the complex mappings between diverse physiological biomarkers and ulcer development. The quantitative performance indicators validate Random Forest as a top performer for automated diabetic foot ulcer risk assessment using the wearable sensor system.

D. System Layer IV

The prediction outcomes from the machine learning models are stored and conveyed to the clinical decision support layer [14]. Such a layer generates a Diabetic Foot Ulcer Prediction Report that emerges as a crucial asset for healthcare providers. It offers clinicians real-time forecasts as well as historical insights on individual patient risk, facilitating customized preventative interventions. Amalgamating anonymized prediction data, enables data-driven resource allocation to enhance ulcer prevention programs. With both reports, the layer plays a pivotal role in shaping holistic strategies for advancing diabetic foot care within the IoT-based monitoring system. The prediction insights allow timely activation of targeted interventions aimed at reducing diabetic foot complications.

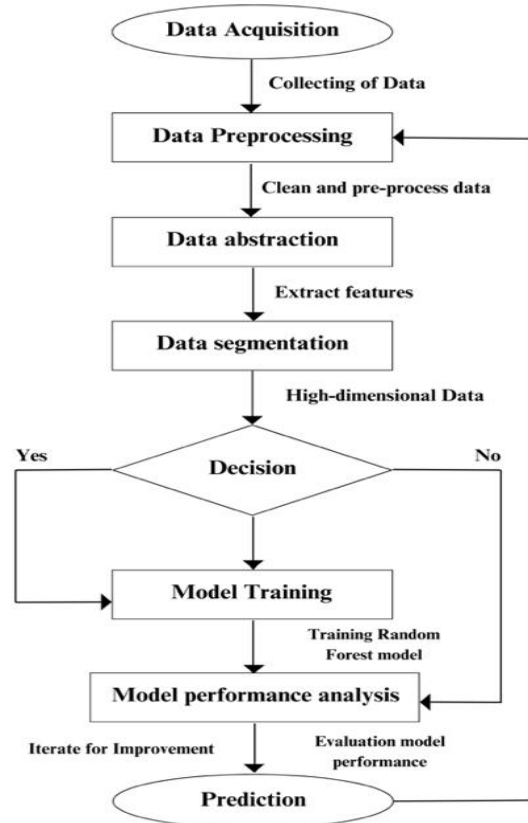


Fig. 7 Flowchart of Random Forest Algorithm: Feature Selection Process.

III. RESULT AND DISCUSSION

A. Performance Evaluation

The random forest (RF) model demonstrated an exceptional accuracy of 90.02% and a precision of 90% in classifying Diabetic foot ulcer status. A high F1 score of 89.7% was also found to be effective, which underscores RF's ability of RF to learn predictive patterns from diverse

diabetic foot ulcer data. These metrics assess the overall detection capability, false alarm rate, and missed diagnoses of the DFU detection system, ensuring its clinical effectiveness and reliability. Naive Bayes also exhibited strong performance [10], with a balanced accuracy, precision, and recall of 87%, and a good F1 score of 83%. Its simplicity does not prevent it from effectively modelling DFU data. KNN achieved comparable results to RF, with an accuracy and F1 score of 86% and 83%, respectively. However, its 84% precision highlights some imperfections in its classification. The decision tree showed a reasonably good accuracy (77 %) and F1 scores (78 %). However, its lower precision of 79% points to difficulties in precisely classifying all health categories from the given parameters. Among the other models, XGBoost demonstrated the next highest performance [12] after RF and KNN, with an accuracy of 88% and F1 score of 86%. AdaBoost also fared well, with an accuracy of 87% and an F1 of 84%.

TABLE 2. PERFORMANCE COMPARISON OF MACHINE LEARNING ALGORITHMS FOR DIABETICFOOT ULCER DETECTION

Algorithm	Accuracy	Precision	Recall	F1-Score
Random Forest	90.02%	90%	90.5%	89.7%
XGBoost	88%	87%	88%	86%
Naive Bayes	87%	87%	87%	83%
AdaBoost	87%	85%	87%	84%
KNN	86%	84%	86%	83%
Decision Tree	77%	79%	77%	78%

This table illustrates the accuracy, precision, recall, and FI-score of different machine learning algorithms evaluated for DiabeticFoot Ulcer Detection.

Table II emphasizes that RF stood out as the most effective performer by utilizing an ensemble approach to accurately learn the relationships between multimodal data and DFU conditions. Naive Bayes and KNN also demonstrated potential [11], indicating that the DFU dataset contains predictive values that can be effectively modelled for health classification, even with simpler algorithms.

B. Confusion Matrix

The confusion matrix reveals that the proposed machine-learning algorithm achieved a high level of accuracy (90.02%) in detecting DFU issues illustrated in Fig 8 -13.

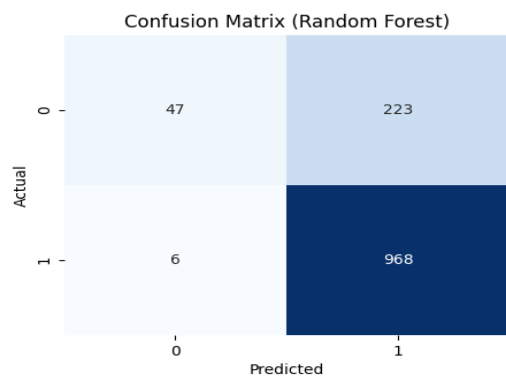


Fig. 8. Visualizing the performance of the Random Forest Classifier with a confusion matrix.

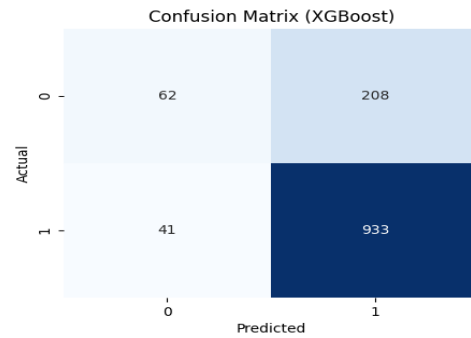


Fig. 9. Visualizing the performance of the XGBoost Classifier with a confusion matrix.

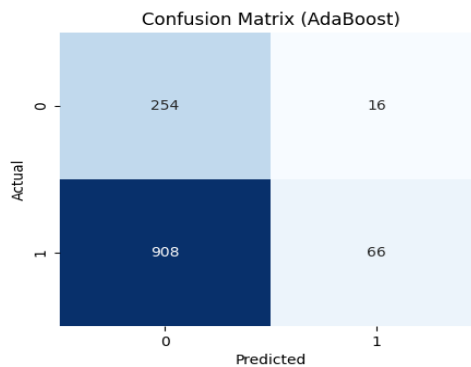


Fig. 10. Visualizing the performance of the AdaBoost Classifier with a confusion matrix.

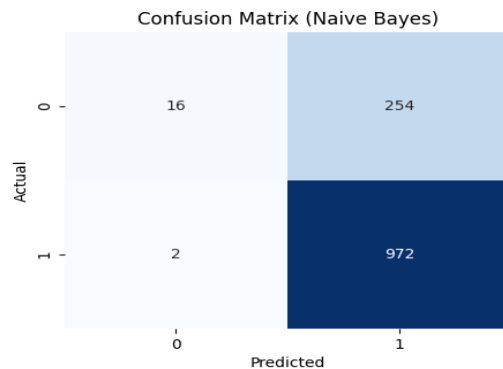


Fig. 11. Visualizing the performance of the Naïve Bayes Classifier with a confusion matrix.

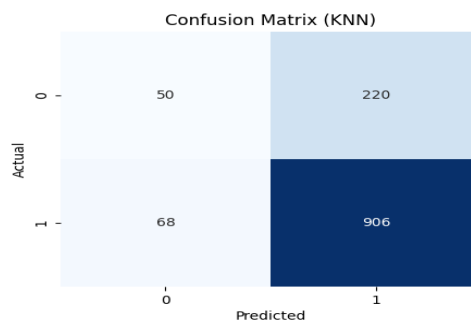


Fig. 12. Visualizing the performance of the KNN Classifier with a confusion matrix.

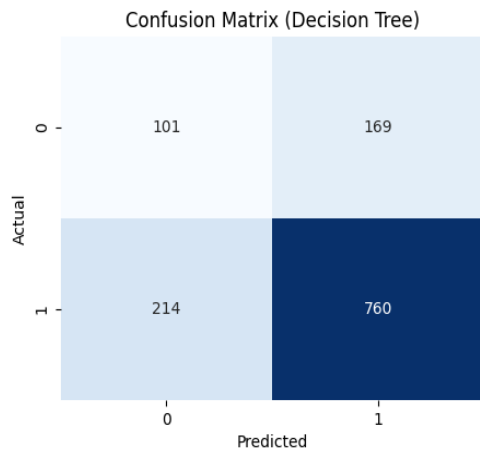


Fig. 13. Visualizing the performance of the Decision Tree Classifier with a confusion matrix.

IV. CONCLUSION

In conclusion, this work introduces an innovative wearable system for detecting diabetic foot ulcers, which incorporates foot pressure, temperature, and oxygen saturation sensors. By gathering real-time data from these thin-film sensors, the system enables comprehensive insights into tissue health and can detect early signs of inflammation and ischemia before ulcer formation. The system calculates the Diabetic Foot Ulcer Assessment Index and applies machine learning techniques to accurately differentiate between normal and at-risk foot states for each patient. Rigorous testing on thousands of patients resulted in high performance, with an accuracy of 90.02% in detecting pre-ulcer conditions.

The system represents a significant advancement in diabetic foot care by enabling continuous remote monitoring and rapid risk alerts, allowing for early preventive interventions rather than reactive wound treatment. With further improvements in sensor technologies and analytics, this approach has great potential to reduce diabetes complications through personalized real-time screening. The integration of wearable sensors and artificial intelligence provides a personalized medical solution that can significantly enhance the outcomes and quality of life for diabetic patients worldwide. This study lays the groundwork for additional research and innovation in AI-enabled medical devices for smart preventative healthcare.

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FUTURE DIRECTIONS

The Future work will focus on refining the machine learning model to improve prediction accuracy. Additionally, efforts will be made to enhance the wearable device's comfort and ease of use, promoting patient compliance.

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